Brief for Alex Gates — Project status, model duality, and path ahead

Date: 2026-06-11
From: Charles Levine (draft for your edit)
Planning context: Cross-project review via **COMPASS** agent; domain agents **CODA** (Army), **SCOUT** (basketball), **PEER** (tenure), **VECTOR** (manuscript/theory).

Purpose: Keep you aligned on where the dissertation program stands, the main modeling tension, and the sequencing plan for a **Summer–Fall 2026** core manuscript draft/submission.

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1. Where we stand (one paragraph)

We have a **credible three-setting empirical story**: inverted-U in advancement vs **leave-one-out (LOO) peer-pool quality** in Army (anchor) and college basketball (replication), with academia (tenure) as a **preliminary third panel (stage 9 bins)**. The scientific bottleneck is no longer "find the curve" but **converge on claim language**, **minimal generative mechanism**, and **2–3 testable predictions** for a Wang-style paper — without letting Army 525/UIC work, network extensions, or generative perfectionism block the draft.

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2. What is solid

Area	Status
Empirical inverted-U (LOO axis)	Army ✅ · Basketball ✅ (`poolq_loo` / draft rate) · Tenure ⚠️ preliminary
Wang empirical ladder (basketball)	Bins → LPM → logit in `538`; L^* reported
Minimal score equation	$S_i = A_i - \lambda \cdot L_{C,\mathrm{LOO}}$ implemented in `538D` CELL 10
Ability-only generative null	Confirmed — congestion term needed for nonlinear readout
Generative POC (one axis)	Inverted-U vs **pool mean** (`team_mean`, includes self) under current top-K knobs
Tenure measurement	End-to-end stage 9 on existing data per your April direction; sample loss logged

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3. The model duality (core tension)

Two related programs have been running in parallel. They are **not** the same estimand unless we nest them explicitly.

A. Empirical stylized fact (cross-domain)

- **Axis: LOO pool quality** — mean teammate performance **excluding self** (`poolq_loo` in basketball; analogous objects in Army and tenure).
- **Finding:** Draft / promotion / tenure rates rise through mid-quality peer environments, then fall in the elite tier (**inverted-U**).
- **Role in paper:** Wang **Rung 1** — the replicated phenomenon.

B. Generative minimal model (Alex score + assignment)

- **Score:** $S_i = A_i - \lambda \cdot L_{C,\mathrm{LOO}}$ — own ability minus **LOO congestion** (viable peers excluding self).
- **Current readout:** Inverted-U appears when success is conditioned on **pool mean** (whole-roster average, **not** LOO).
- **On the empirical axis:** With the same knobs, generative readout vs **LOO pool quality** is **mostly decreasing**, not inverted-U.
- **Role in paper:** Wang **Rung 2** — proof that finite distinction / congestion can bend advancement; **partial** match to Rung 1 unless axis language is careful.

C. Decomposition / predictions track (Charles original direction)

- **Structure:** Local environment as one object: $L_{\mathrm{net}} = B(\cdot) - D(\cdot)$ — **benefit (development)** minus **constraint (congestion / distinction cost)**.
- **Empirical mechanism columns** (`congestion_quality`, LOO dispersion, minutes, etc.) intended as **diagnostics of B vs D**, not a second mechanism.
- **Risk:** Prior guidance sounded like "**Alex model for curves, multivariate empirical for predictions**" — scientifically workable only if nested in **one** ontology; otherwise it reads as **two models**.

D. Terminology lock (to avoid axis confusion)

Quantity	Name rule
Excluding self	Must say **LOO** (e.g. LOO pool quality, LOO congestion)
Whole roster including self	**Pool mean** / `team_mean` — **not** LOO

Bottom line for you: The program is **not** stuck on whether the empirical U exists. It is stuck on whether the **generative object**, the **empirical decomposition**, and the **conditioning axis** are **one scientific story** or two parallel tracks — and on how honest the manuscript can be about that gap on the LOO axis.

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4. Scientific end state vs near-term plan

What we think is scientifically strongest (longer run)

A **single decomposable generative model** that:

1. Keeps the Alex score as the **constraint leg** (D) or equivalent,
2. Adds an explicit **benefit / exposure** channel (B) where needed,
3. Reproduces an inverted-U when conditioned on **LOO pool quality** — the same axis as the empirical stylized fact.

That is the clean Wang ladder: Rung 2 **explains** Rung 1 on the **same** estimand.

What we locked for Summer–Fall 2026 (near-term)

Manuscript-first path — draft now with **honest layering**, not a hard gate on generative LOO-pool-quality bin-for-bin match:

Layer	Manuscript role	Blocks draft?
Empirical LOO inverted-U (3 settings)	Main empirical contribution	No
Alex score + assignment (pool-mean readout)	Minimal generative POC + **explicit two-row axis table**	No
B–D nesting prose + mechanism-column predictions	§3 mechanism + §4 predictions	Yes — needs SCOUT/VECTOR write-up
Generative match on LOO pool quality	Ideal upgrade; parallel science	**No**

Parallel track (recommended, time-boxed): SCOUT attempts **one nesting milestone** toward unified LOO generative readout (e.g. explicit B term or assignment-rule fix). If it lands before submission, we **upgrade** §3; if not, we ship with limitations — not restart the whole timeline.

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5. Sequenced work (next ~6–10 weeks)

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Week 1–2   Lock claim language · SCOUT exports (empirical Fig 2 + generative + axis table)
           · PEER Cell 12 Cox in parallel (soft gate — not blocking first VECTOR draft)
           · CODA manuscript figures + estimand-checked captions

Week 2–4   VECTOR drafts Wang-structure manuscript (empirical triad → minimal model → predictions)
           · SCOUT answers nesting questions (see §6)

Parallel   Optional 3–4 week SCOUT sprint on LOO generative nesting (Path I milestone)
           · Tenure Cox table before submission
           · Defer: Army 525/TB-stratify, network extensions, full HPC sweeps, Fine-Gray
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Target: Core manuscript draft/submission **Summer–Fall 2026**.

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6. What we are asking SCOUT (routing via Charles)

Primary doc: `3-Master_Plan/20260611_0911_COMPASS_to_SCOUT_model_coherence_questions.md`

We need SCOUT to supply:

1. **Nesting chain** — where the Alex score lives in $L_{\text{net}} = B - D$; mechanism-column map to B vs D.
2. **Exact sentence** VECTOR may use in §3 so reviewers do not read "two mechanisms."
3. **Deliverables** — axis table, frozen score equation, figure paths, one prediction from the same story.
4. **Single next coding task** under the locked manuscript path (not an open-ended rebuild first).

Your input on whether this ask matches your April–May memos (theory ≠ minimal model; quadratic diagnostic; assortative pools + local comparison) would be valuable.

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7. Tenure (Setting 3) — aligned with your direction

- **Stage 9 binned figure** + honest limitations: sufficient for VECTOR to **start** Setting 3 prose.
- **One basic Cell 12 Cox run** (mirror Army: z-scored LOO pool quality, quadratic, own-performance control) — **parallel before submission**, not a pre-draft blocker.
- **Fine–Gray:** deferred.
- Follows your "end-to-end on existing data, log sample loss, then move on" guidance.

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8. Questions for you (optional — reply in any form)

1. **Generative priority:** For the core paper, is **honest partial POC on pool mean** + **empirical LOO U** acceptable for v1, with LOO generative match as revision/supplement — or would you elevate LOO generative match to a **pre-submission** requirement?

2. **Two-track framing:** Is "score equation for generative curve; decomposed empirical for predictions" still your operational advice? If yes, how would you **nest** it in one Wang object for the manuscript?

3. B **term:** Do you see a minimal **development / exposure** channel as required in the generative sim to hit the LOO axis — or is the pool-mean vs LOO gap fundamentally about **conditioning / assignment estimand**?

4. **Tenure Cox:** Confirm soft gate (stage 9 + one Cox pass) is enough for a three-setting draft you would stand behind.

5. **Near-term cuts:** Anything in §5 you would reprioritize (e.g. CELL 7 robustness, CELL 4D heterogeneity, Army TB-stratify)?

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9. Key repo pointers

Doc	Content
`3-Master_Plan/PROJECT_STATUS_AND_NEAR_TERM_PLAN.md`	Full near-term plan
`3-Master_Plan/20260611_COMPASS_Initial_Review.md`	Structured situational review
`3-Master_Plan/SCOUT_report_to_COMPASS.md`	Basketball / generative status
`3-Master_Plan/PEER_report_to_COMPASS.md`	Tenure status
`3-Master_Plan/CODA_report_to_COMPASS.md`	Army status
`sports/documents/538_Cell10_Generative_Manual.md`	CELL 10 implementation reference

Draft for Charles review and edit before sending to Alex.